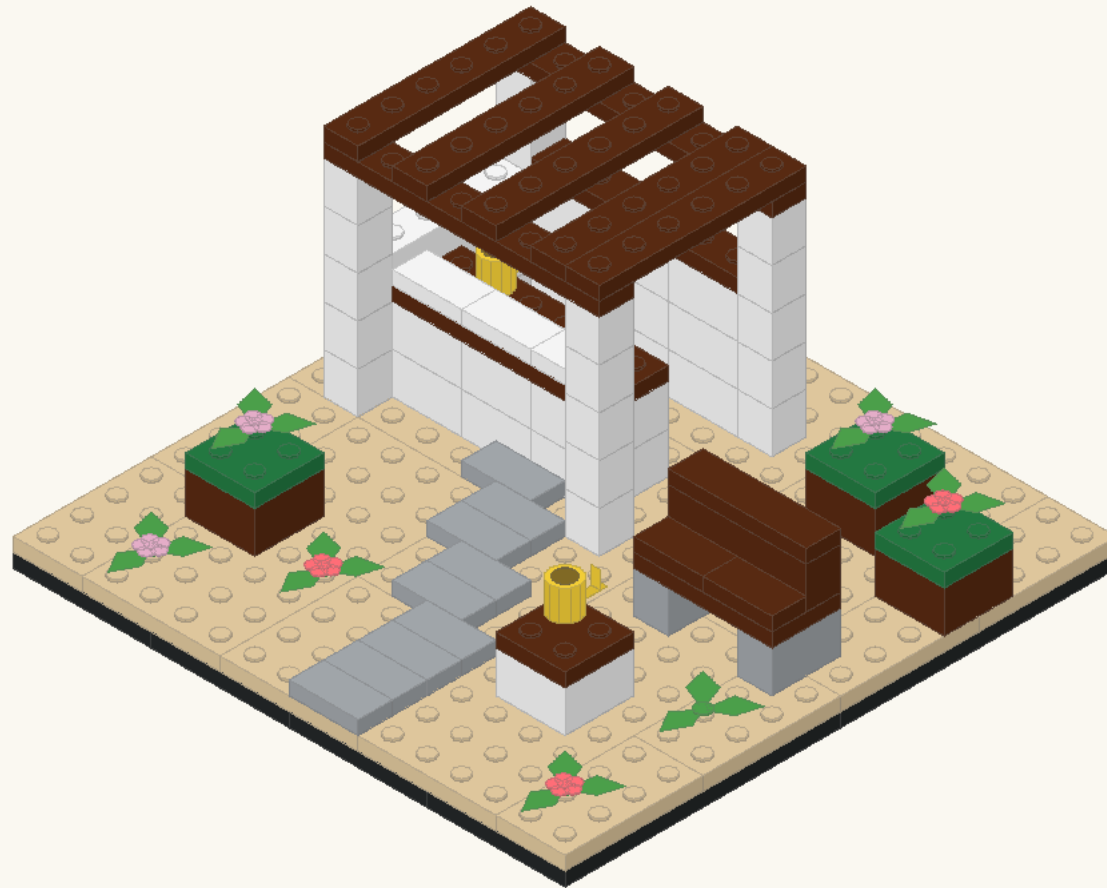


Fern & Steam

A little tea counter, an airy pergola and a courtyard full of plants.



115

PIECES

16 x 16

STUD FOOTPRINT

25

EXACT ELEMENTS

12

BUILD STEPS

Model-derived schematic. Foliage and mugs are simplified. Digital fine-fit and physical build remain to be tested.

What has been checked

Use this as a first physical prototype. A passing inventory calculation is not a completed physical test.

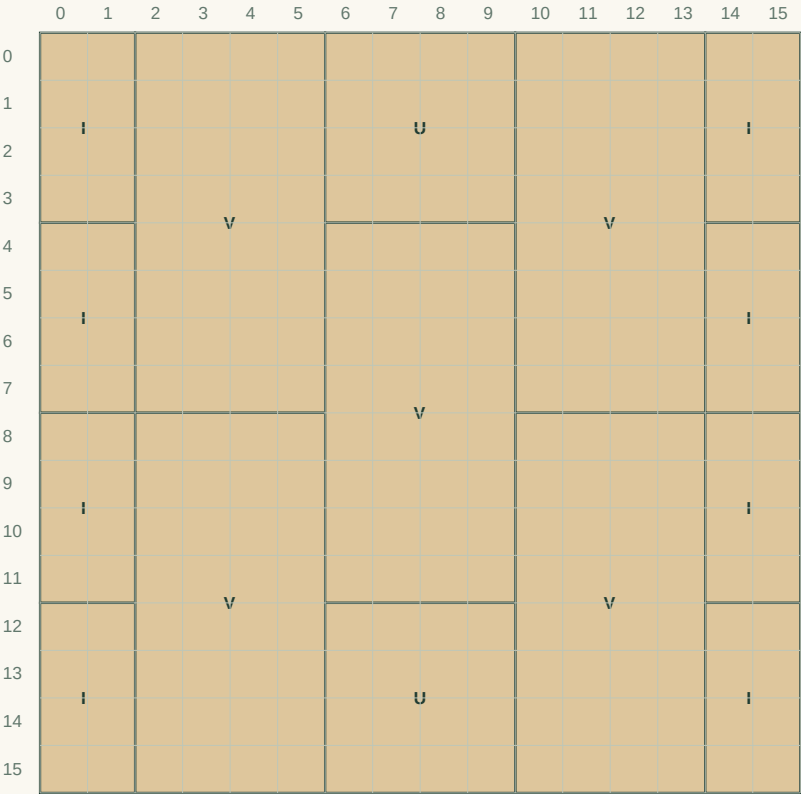
PASS	115 pieces fit the conservative exact-ID inventory.
PASS	Structural boxes have no body overlaps or out-of-bounds placements.
PASS	Every elevated instance has a prior stud support in the simplified model.
PASS	The final simplified connection graph is one connected assembly.
PASS	No earlier structural box blocks the stated downward placement order.
REVIEW	15 decorative instances use mounting abstractions; detailed mesh fit is not tested.
REVIEW	Element-to-LDraw mappings and cached colors need visual confirmation.
PENDING	Physical fit, stability, clutch strength and a complete trial build.

The donor allocation takes at most one of each listed element from each set copy. It currently spans 26 sets. This is a reproducible conservative allocation, not the smallest possible donor selection. Check familiar sets for extra copies first if you want to reduce disassembly.

Build on a flat surface. Finish the two-layer base before lifting it. Attach the furniture, flowers and mugs before closing the pergola. If a piece does not seat naturally, stop and record the exact instance and step instead of forcing the connection.

A precise position for every piece

Keep the front edge toward you. Each step includes a shaded model, a top view and an ordered placement table.



FRONT

X / Y

X runs left to right. Y runs from the rear toward the front. Both start at 0 and describe the rear-left corner of a part footprint.

Z

Z is the bottom elevation measured in plate heights. A plate is 1 unit; an ordinary brick is 3. The finished tan base surface is Z = 2.

FOOTPRINT

Width x depth describes how the part lies in the top view. For example 8 x 1 means a long horizontal beam; 1 x 6 runs front to back.

KEY / ID

Letters A-Y identify exact elements in the parts palette. P001 etc. identify individual placements. Follow placement rows from top to bottom.

DECORATIONS

Leaf Z is its underside mounting plane. A leaf adds 0.375 plate units before its flower; this is intentional. Leaf/flower shapes in the schematic are simplified.

SHADING

Current-step pieces keep their colors. Earlier pieces are muted. The top view shows the uppermost footprint when several pieces share a position.

Your picking palette

Required quantities are counted from the model. Minimum quantities assume complete, accessible donor sets.

KEY	ELEMENT ID	DESCRIPTION	COLOR	NEED	MINIMUM	LEFT*
 A	300301	Brick 2 x 2	White	1	19	18
 B	300401	Brick 1 x 2	White	15	43	28
 C	300501	Brick 1 x 1	White	20	43	23
 D	301001	Brick 1 x 4	White	6	38	32
 E	302228	Plate 2 x 2	Green	3	5	2
 F	303526	Plate 4 x 8	Black	8	8	0
 G	306901	Tile 1 x 2	White	3	45	42
 H	389924	Mug	Yellow	2	12	10
 I	4114309	Plate 2 x 4	Tan	8	27	19
 J	4211151	Tile 1 x 2	Reddish brown	2	7	5
 K	4211186	Plate 2 x 4	Reddish brown	1	17	16
 L	4211194	Tile 1 x 4	Reddish brown	1	3	2
 M	4211210	Brick 2 x 2	Reddish brown	3	8	5

*Remaining conservative budget after this model. It is not an exact physical stock count. Part color labels are normalized from the repo enrichment cache; physically compare the piece against its exact element ID.

Your picking palette

Required quantities are counted from the model. Minimum quantities assume complete, accessible donor sets.

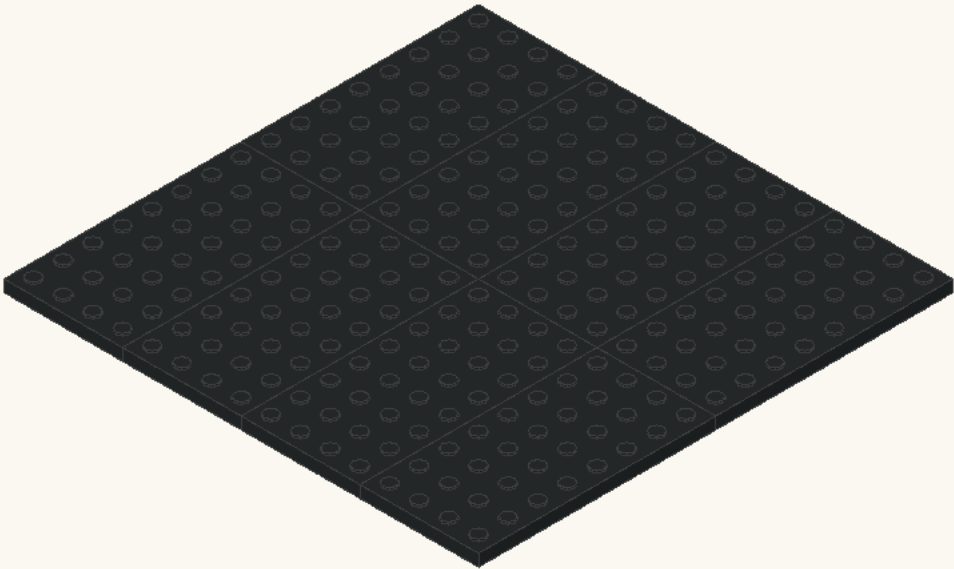
KEY	ELEMENT ID	DESCRIPTION	COLOR	NEED	MINIMUM	LEFT*
 N	4211225	Brick 1 x 4	Reddish brown	1	7	6
 O	4211247	Plate 2 x 6	Reddish brown	1	6	5
 P	4211388	Brick 1 x 2	Light bluish grey	2	19	17
 Q	4211414	Tile 1 x 2	Light bluish grey	9	23	14
 R	4216695	Plate 2 x 2	Reddish brown	1	13	12
 S	4216945	Plate 1 x 8	Reddish brown	2	3	1
 T	4221590	Plate 1 x 6	Reddish brown	6	11	5
 U	4243824	Plate 4 x 4	Tan	2	5	3
 V	4509897	Plate 4 x 8	Tan	5	11	6
 W	6182261	Plant with three leaves	Bright green	7	28	21
 X	6212501	Flower with five petals	Bright pink	3	13	10
 Y	6295242	Flower with five petals	Coral	3	13	10

*Remaining conservative budget after this model. It is not an exact physical stock count. Part color labels are normalized from the repo enrichment cache; physically compare the piece against its exact element ID.

Lay out the foundation

On a flat surface, lay the eight black plates edge to edge. They are loose until the next step; do not lift them yet.

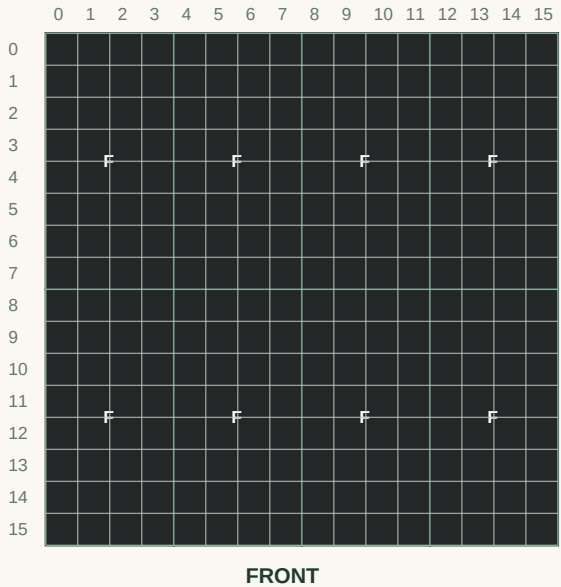
INSTANCE	KEY	X	Y	Z	FOOTPRINT
P001	F	0	0	0	4 x 8
P002	F	4	0	0	4 x 8
P003	F	8	0	0	4 x 8
P004	F	12	0	0	4 x 8
P005	F	0	8	0	4 x 8
P006	F	4	8	0	4 x 8
P007	F	8	8	0	4 x 8
P008	F	12	8	0	4 x 8



ADD 8 PIECES / TOTAL 8

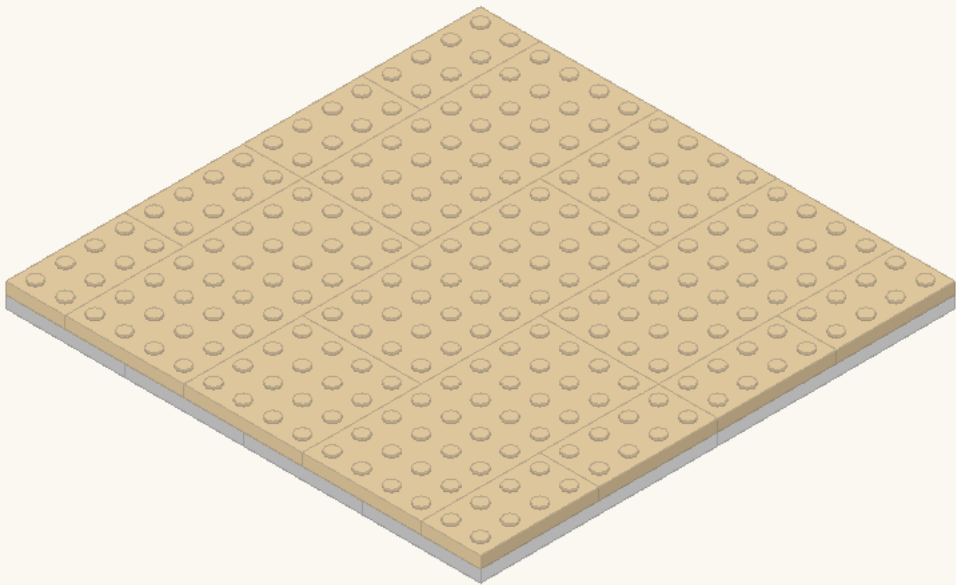
F x8

Assembly schematic; use the coordinates for exact placement.



Lock the base together

Press the tan layer onto the black layer. The offset seams join the foundation into a single assembly. Finish all 15 tan plates before lifting.

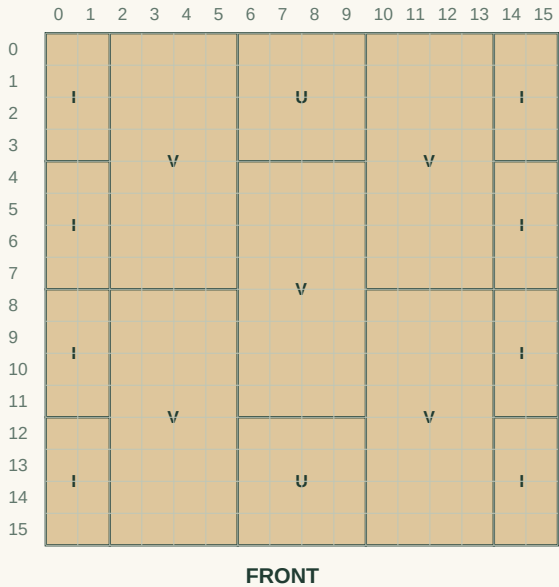


ADD 15 PIECES / TOTAL 23

I x8 U x2 V x5

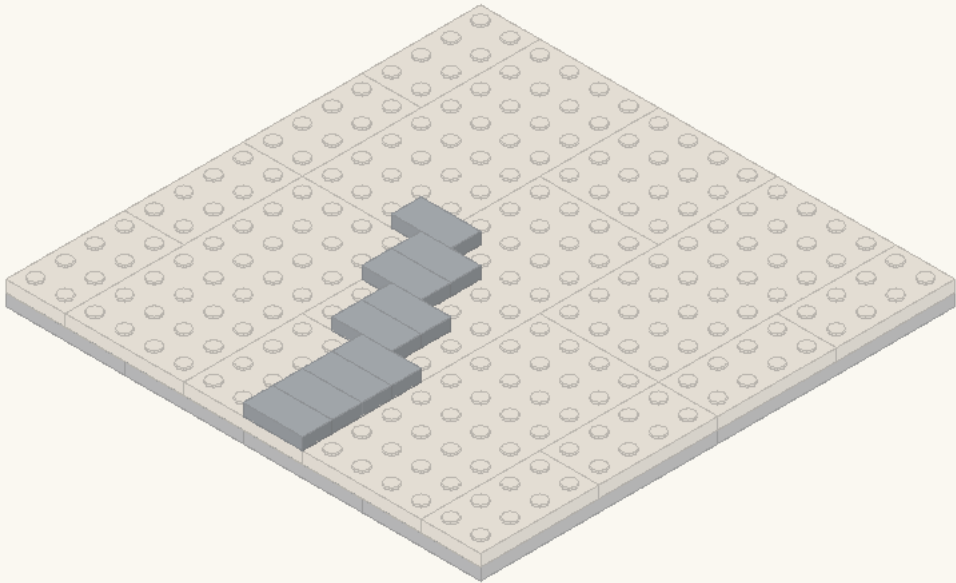
Assembly schematic; use the coordinates for exact placement.

INSTANCE	KEY	X	Y	Z	FOOTPRINT
P009	V	2	0	1	4 x 8
P010	V	2	8	1	4 x 8
P011	V	10	0	1	4 x 8
P012	V	10	8	1	4 x 8
P013	U	6	0	1	4 x 4
P014	V	6	4	1	4 x 8
P015	U	6	12	1	4 x 4
P016	I	0	0	1	2 x 4
P017	I	0	4	1	2 x 4
P018	I	0	8	1	2 x 4
P019	I	0	12	1	2 x 4
P020	I	14	0	1	2 x 4
P021	I	14	4	1	2 x 4
P022	I	14	8	1	2 x 4
P023	I	14	12	1	2 x 4



Lay the courtyard path

Lay the light grey tiles in a stepped path from the front entrance toward the tea counter. Keep the marked areas studded for furniture and plants.

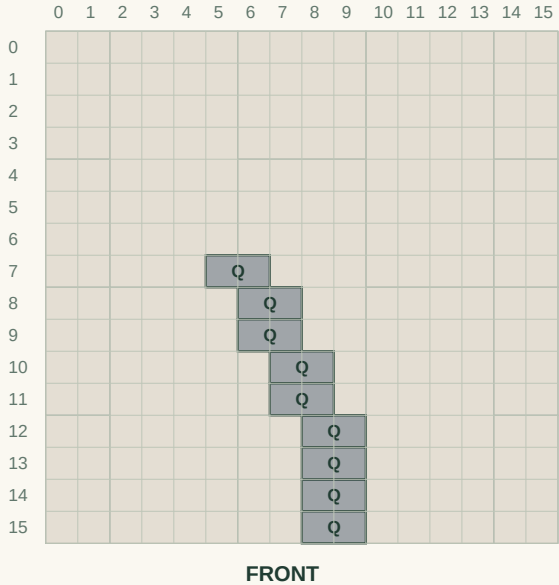


ADD 9 PIECES / TOTAL 32

Q x9

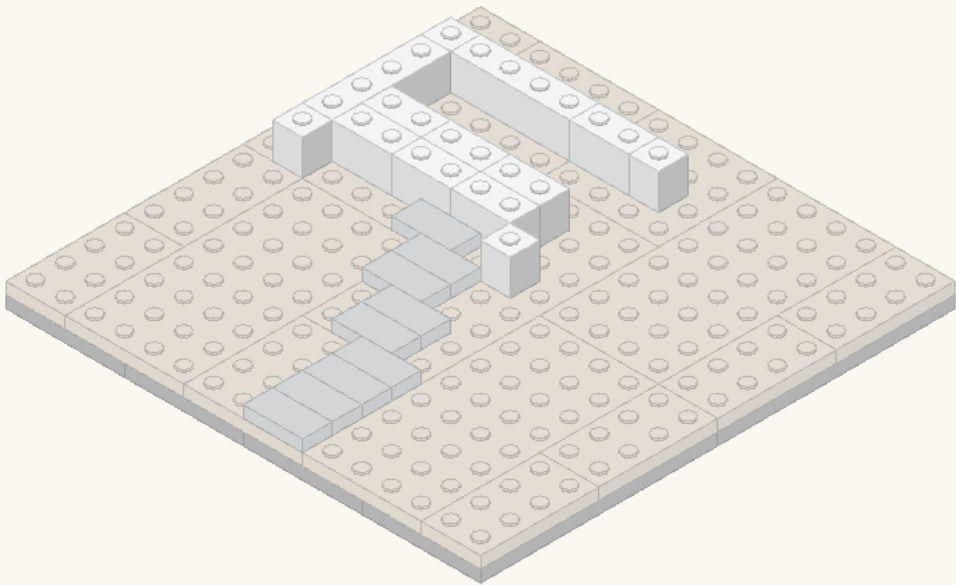
Assembly schematic; use the coordinates for exact placement.

INSTANCE	KEY	X	Y	Z	FOOTPRINT
P024	Q	8	14	2	2 x 1
P025	Q	8	15	2	2 x 1
P026	Q	8	12	2	2 x 1
P027	Q	8	13	2	2 x 1
P028	Q	7	10	2	2 x 1
P029	Q	7	11	2	2 x 1
P030	Q	6	8	2	2 x 1
P031	Q	6	9	2	2 x 1
P032	Q	5	7	2	2 x 1



Raise the kiosk, course 1

Add the four white corner posts, low back/side walls and the first course of the tea counter. The open side faces the path.

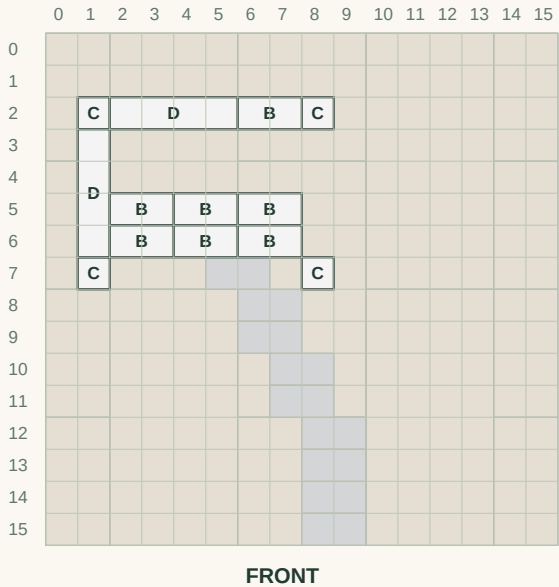


INSTANCE	KEY	X	Y	Z	FOOTPRINT
P033	C	1	2	2	1 x 1
P034	C	8	2	2	1 x 1
P035	C	1	7	2	1 x 1
P036	C	8	7	2	1 x 1
P037	D	2	2	2	4 x 1
P038	B	6	2	2	2 x 1
P039	D	1	3	2	1 x 4
P040	B	2	5	2	2 x 1
P041	B	4	5	2	2 x 1
P042	B	6	5	2	2 x 1
P043	B	2	6	2	2 x 1
P044	B	4	6	2	2 x 1
P045	B	6	6	2	2 x 1

ADD 13 PIECES / TOTAL 45

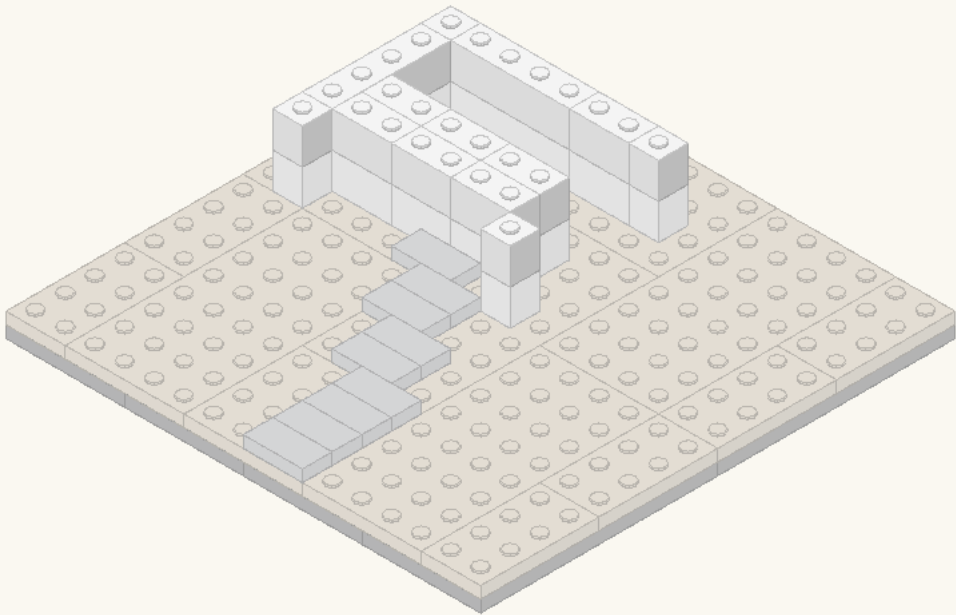
B x7 C x4 D x2

Assembly schematic; use the coordinates for exact placement.



Raise the kiosk, course 2

Repeat the post and wall positions one brick higher; complete the second and final course of the tea counter.

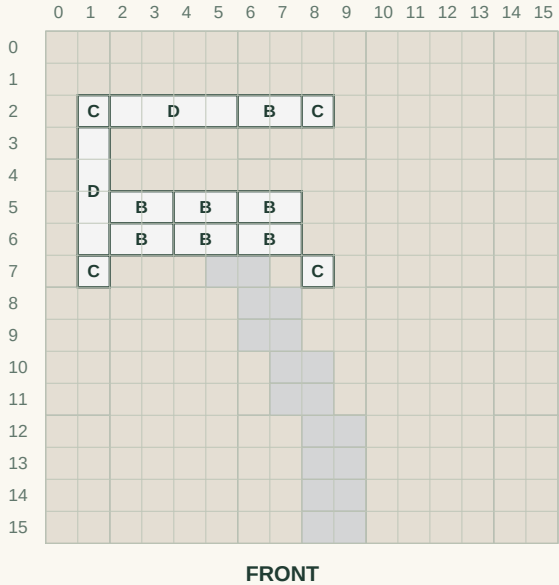


ADD 13 PIECES / TOTAL 58

B x7 C x4 D x2

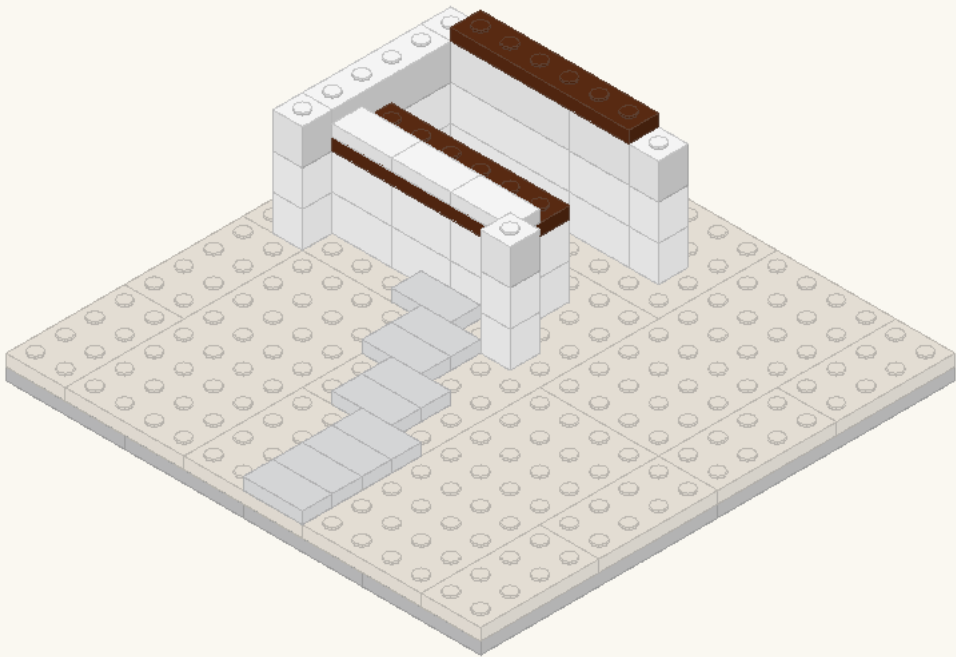
Assembly schematic; use the coordinates for exact placement.

INSTANCE	KEY	X	Y	Z	FOOTPRINT
P046	C	1	2	5	1 x 1
P047	C	8	2	5	1 x 1
P048	C	1	7	5	1 x 1
P049	C	8	7	5	1 x 1
P050	D	2	2	5	4 x 1
P051	B	6	2	5	2 x 1
P052	D	1	3	5	1 x 4
P053	B	2	5	5	2 x 1
P054	B	4	5	5	2 x 1
P055	B	6	5	5	2 x 1
P056	B	2	6	5	2 x 1
P057	B	4	6	5	2 x 1
P058	B	6	6	5	2 x 1



Finish the wall and counter

Raise the posts and back/side walls one more course. Add the brown counter plate, the narrow back shelf, and the white counter trim.

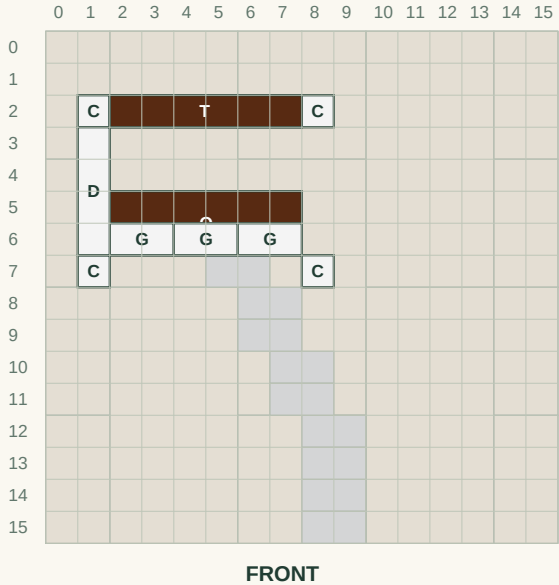


ADD 12 PIECES / TOTAL 70

B x1 C x4 D x2 G x3 O x1 T x1

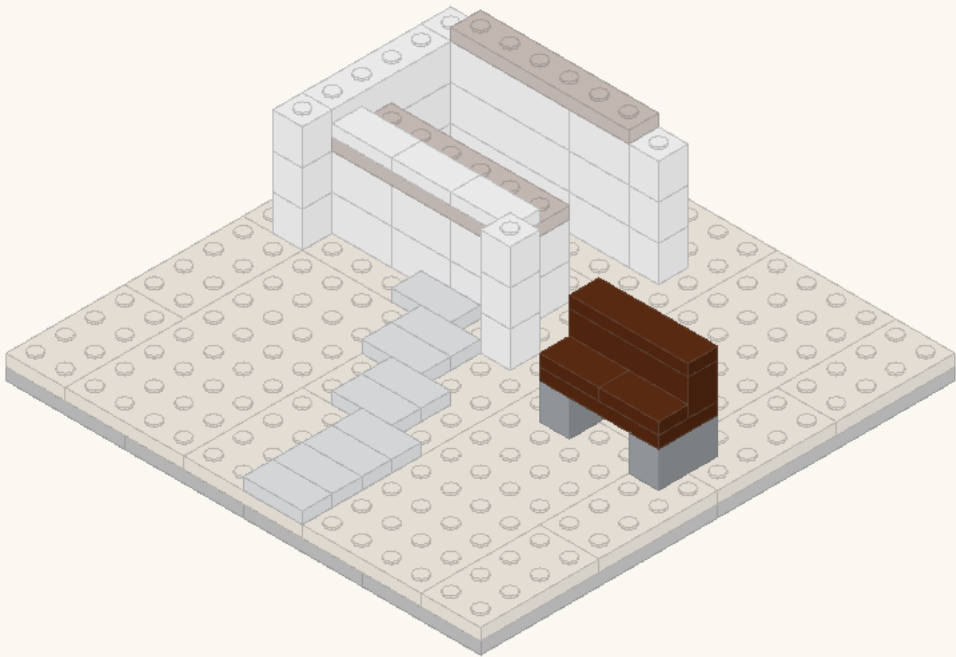
Assembly schematic; use the coordinates for exact placement.

INSTANCE	KEY	X	Y	Z	FOOTPRINT
P059	C	1	2	8	1 x 1
P060	C	8	2	8	1 x 1
P061	C	1	7	8	1 x 1
P062	C	8	7	8	1 x 1
P063	D	2	2	8	4 x 1
P064	B	6	2	8	2 x 1
P065	D	1	3	8	1 x 4
P066	O	2	5	8	6 x 2
P067	T	2	2	11	6 x 1
P068	G	2	6	9	2 x 1
P069	G	4	6	9	2 x 1
P070	G	6	6	9	2 x 1



Make the courtyard bench

Add two grey supports, the brown seat plate, the front seat tiles and the upright backrest. The bench faces the front edge.

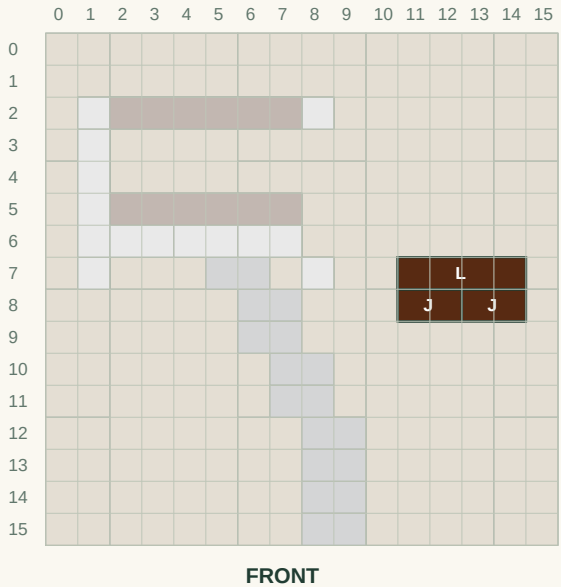


INSTANCE	KEY	X	Y	Z	FOOTPRINT
P071	P	11	7	2	1 x 2
P072	P	14	7	2	1 x 2
P073	K	11	7	5	4 x 2
P074	J	11	8	6	2 x 1
P075	J	13	8	6	2 x 1
P076	N	11	7	6	4 x 1
P077	L	11	7	9	4 x 1

ADD 7 PIECES / TOTAL 77

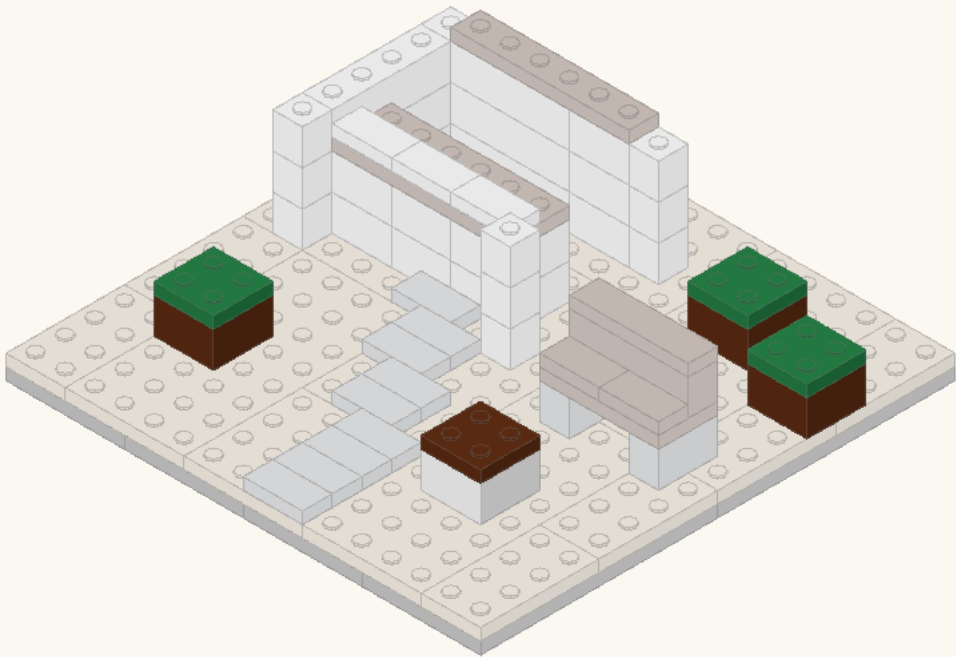
J x2 K x1 L x1 N x1 P x2

Assembly schematic; use the coordinates for exact placement.



Add table and planters

Add the small white table pedestal with brown top, then three brown planters capped with green plates.

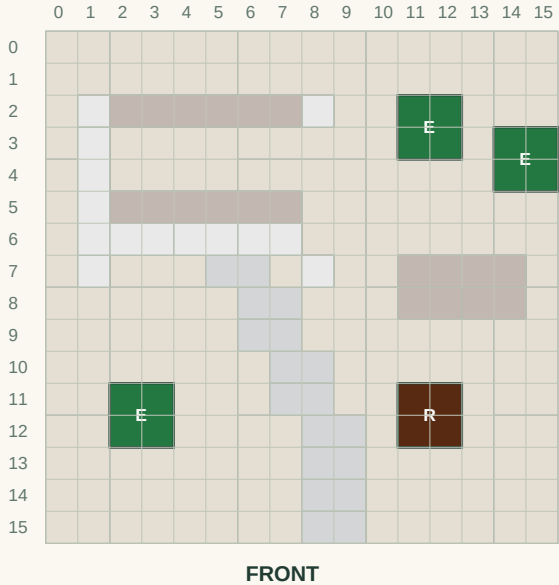


ADD 8 PIECES / TOTAL 85

A x1 E x3 M x3 R x1

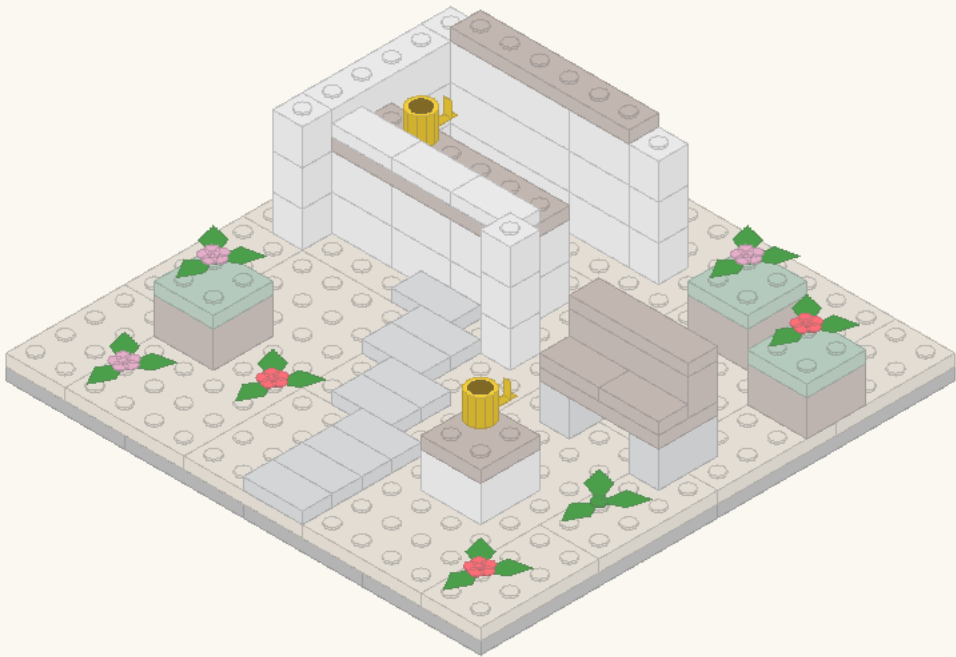
Assembly schematic; use the coordinates for exact placement.

INSTANCE	KEY	X	Y	Z	FOOTPRINT
P078	A	11	11	2	2 x 2
P079	R	11	11	5	2 x 2
P080	M	11	2	2	2 x 2
P081	E	11	2	5	2 x 2
P082	M	14	3	2	2 x 2
P083	E	14	3	5	2 x 2
P084	M	2	11	2	2 x 2
P085	E	2	11	5	2 x 2



Plant the garden and serve tea

Attach the leaves and flowers at the marked studs. BOTH mug handles point REARWARD: the counter mug clears the white trim, and the table mug clears its neighboring stud.

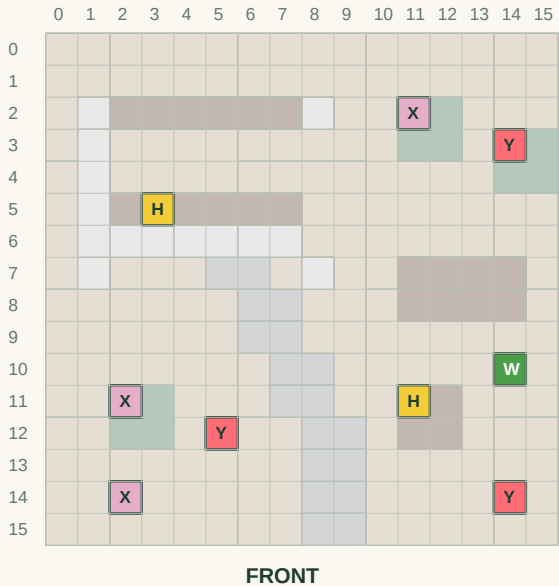


ADD 15 PIECES / TOTAL 100

H x2 W x7 X x3 Y x3

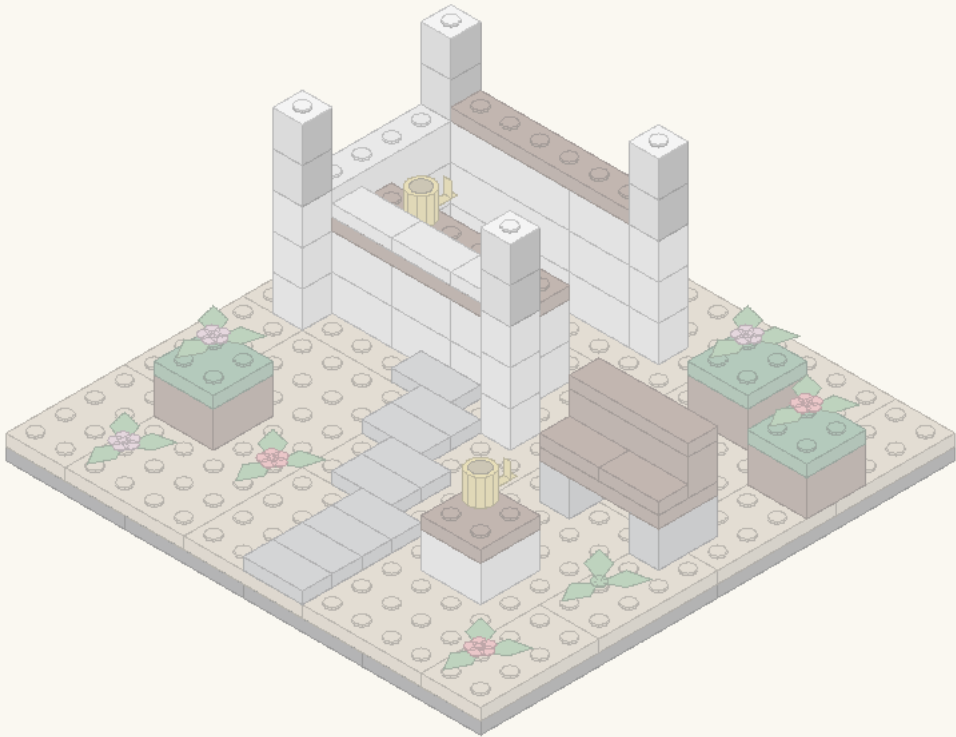
Assembly schematic; use the coordinates for exact placement.

INSTANCE	KEY	X	Y	Z	FOOTPRINT
P086	W	11	2	6	1 x 1
P087	X	11	2	6.375	1 x 1
P088	W	14	3	6	1 x 1
P089	Y	14	3	6.375	1 x 1
P090	W	2	11	6	1 x 1
P091	X	2	11	6.375	1 x 1
P092	W	5	12	2	1 x 1
P093	Y	5	12	2.375	1 x 1
P094	W	2	14	2	1 x 1
P095	X	2	14	2.375	1 x 1
P096	W	14	14	2	1 x 1
P097	Y	14	14	2.375	1 x 1
P098	W	14	10	2	1 x 1
P099	H	3	5	9	1 x 1
P100	H	11	11	6	1 x 1



Raise the pergola posts

Add two more white bricks to each of the four posts. Complete all four posts to the same height.

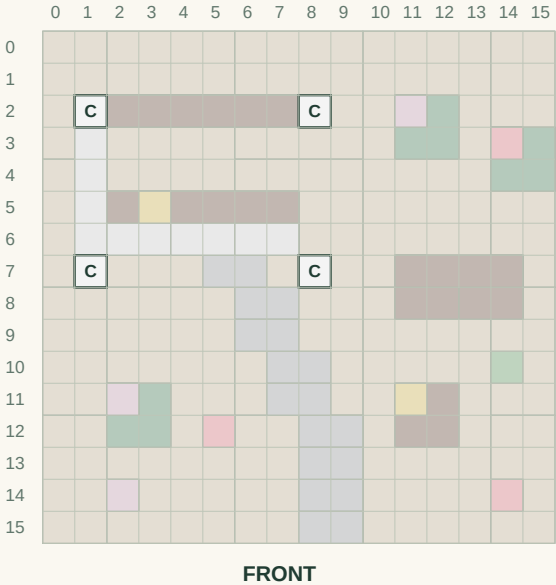


ADD 8 PIECES / TOTAL 108

C x8

Assembly schematic; use the coordinates for exact placement.

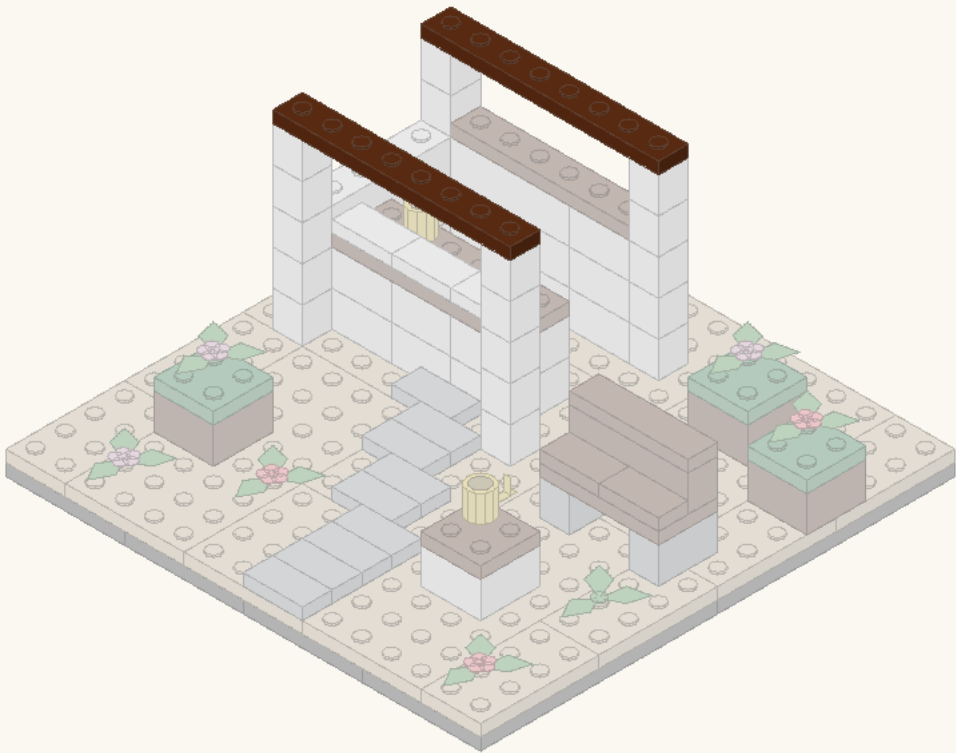
INSTANCE	KEY	X	Y	Z	FOOTPRINT
P101	C	1	2	11	1 x 1
P102	C	8	2	11	1 x 1
P103	C	1	7	11	1 x 1
P104	C	8	7	11	1 x 1
P105	C	1	2	14	1 x 1
P106	C	8	2	14	1 x 1
P107	C	1	7	14	1 x 1
P108	C	8	7	14	1 x 1



Fit the pergola beams

Fit the two brown 1 x 8 plates left-to-right across the front and rear pairs of posts. Support the posts while pressing the beam ends.

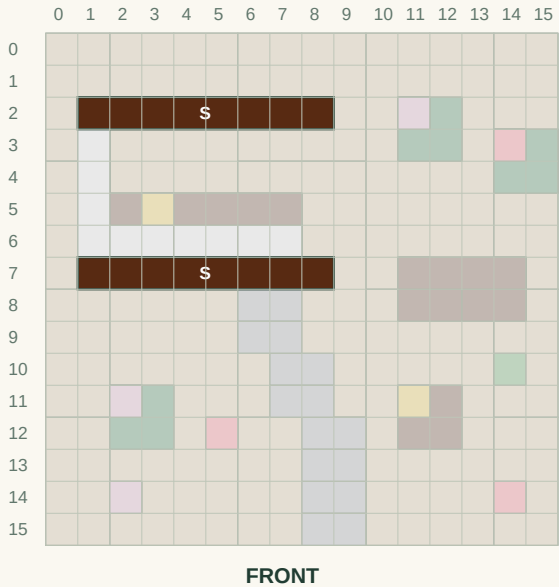
INSTANCE	KEY	X	Y	Z	FOOTPRINT	
P109	S		1	2	17	8 x 1
P110	S		1	7	17	8 x 1



ADD 2 PIECES / TOTAL 110

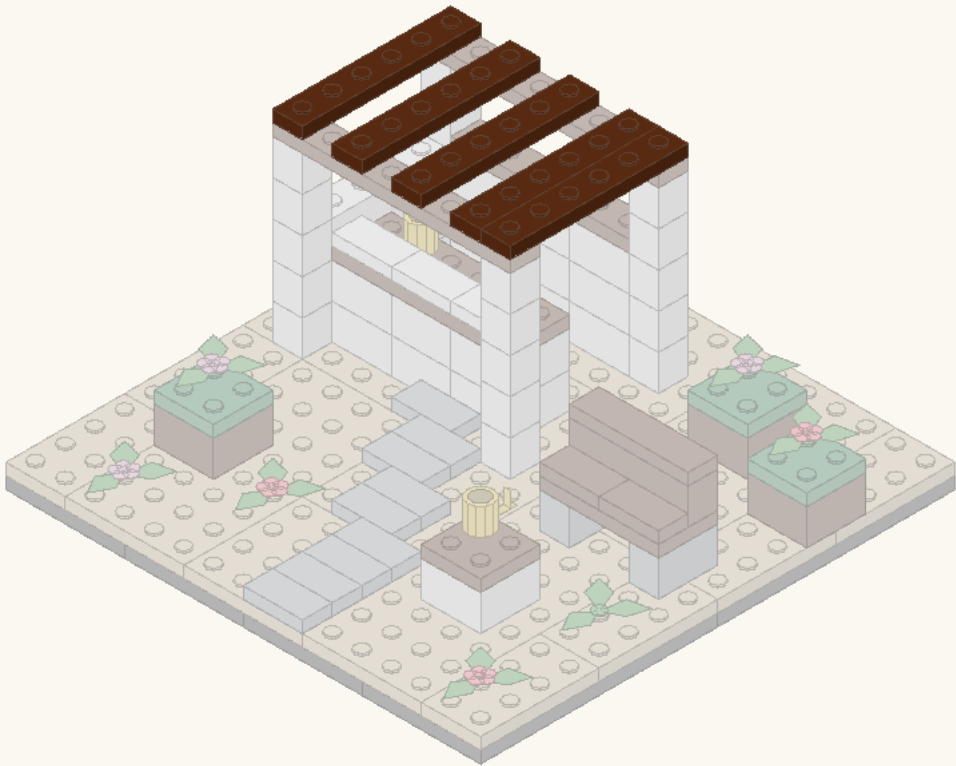
S x2

Assembly schematic; use the coordinates for exact placement.



Fit the five roof slats

Lay five brown 1 x 6 plates front-to-back across both beams. Keep the intentional gaps between slats. Review the completed model before lifting it by its base.

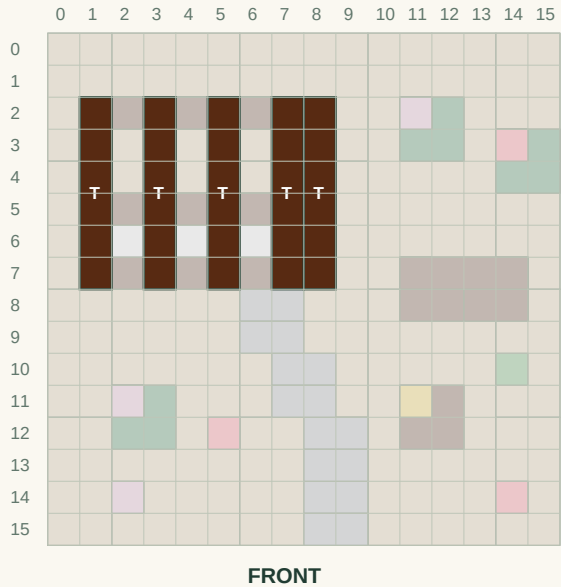


INSTANCE	KEY	X	Y	Z	FOOTPRINT	
P111	T		1	2	18	1 x 6
P112	T		3	2	18	1 x 6
P113	T		5	2	18	1 x 6
P114	T		7	2	18	1 x 6
P115	T		8	2	18	1 x 6

ADD 5 PIECES / TOTAL 115

T x5

Assembly schematic; use the coordinates for exact placement.



Record the physical result

This revision is ready for a careful trial assembly. Keep a short record so the next revision improves on evidence.

CHECK	RESULT / NOTES
All exact elements picked and visually matched	
Two-layer base holds together when lifted	
Counter, wall and posts seat without forcing	
Bench seat and backrest attach securely	
Leaf-to-flower connections seat correctly	
Mug handles clear surrounding pieces	
Both beams connect to all four posts	
Slats fit without bending posts or beams	
Completed model stays stable on a level surface	

Companion files: fern-and-steam.ldr opens in an LDraw-compatible editor; model.json records every placement; picking-list.md groups the exact IDs by donor set; inventory-snapshot.json and verification.json preserve the inventory assumptions and checks.

Geometry references: library.ldraw.org/library/official/parts/ | File format: ldraw.org/article/218.html